

SOFTWARE REQUIREMENTS SPECIFICATION

AIDCELIX E-Pharmacy Application

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This document specifies the functional and non-functional requirements for the AIDCELIX E-Pharmacy stock checker and medication delivery platform.

TABLE OF CONTENTS

1. Introduction
 - 1.1 Purpose
 - 1.2 Scope
 - 1.3 Definitions, Acronyms, and Abbreviations
 - 1.4 References
 - 1.5 Overview
2. Overall Description
 - 2.1 Product Perspective
 - 2.2 Product Functions
 - 2.3 User Classes and Characteristics
 - 2.4 Operating Environment
 - 2.5 Design and Implementation Constraints
 - 2.6 Assumptions and Dependencies
3. System Features (Functional Requirements)
 - 3.1 User Account Management
 - 3.2 Location-Based Pharmacy Search
 - 3.3 Medication Search and Display
 - 3.4 Medication Ordering and Delivery
 - 3.5 Payment Processing
 - 3.6 Gozem API Integration
 - 3.7 Cross-Platform Compatibility
4. External Interface Requirements
 - 4.1 User Interfaces
 - 4.2 Hardware Interfaces
 - 4.3 Software Interfaces
 - 4.4 Communications Interfaces
5. Non-Functional Requirements
 - 5.1 Performance Requirements
 - 5.2 Security Requirements
 - 5.3 Availability and Reliability

5.4 Scalability

6. Appendices

1. INTRODUCTION

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a detailed description of the functional and non-functional requirements for the AIDCELIX E-Pharmacy Application. This document is intended for use by the development team, project managers, quality assurance team, and stakeholders to ensure a common understanding of the system to be built. It serves as the primary reference for design, development, testing, and acceptance of the AIDCELIX platform.

1.2 Scope

AIDCELIX is an E-Pharmacy stock checker and medication ordering platform designed to connect users with nearby pharmacies. The application enables users to create personal accounts, search for medications with real-time pricing, locate the nearest pharmacies stocking specific drugs, place orders for home delivery, and process payments through integrated mobile money services. The system integrates with the MEDindex Pro pricing database for medication costs and the Gozem delivery service API for logistics fulfillment. AIDCELIX will be accessible on both mobile phones and desktop computers.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
AIDCELIX	The E-Pharmacy stock checker and medication delivery application.
MEDindex Pro	External application/database providing medication pricing data.
Gozem	Third-party delivery service application integrated for medication delivery.
Orange Money	Mobile money payment service provided by Orange Telecom.
MTN Money	Mobile money payment service provided by MTN Group.
API	Application Programming Interface; enables communication between AIDCELIX and Gozem.
E-Pharmacy	Electronic pharmacy platform enabling online medication search and ordering.
User	Any individual who registers and uses the AIDCELIX application.
Command	User action to place an order for medication through the application.

1.4 References

- MEDindex Pro Application — External pricing reference database.
- Gozem Delivery Application — Third-party logistics and delivery service.
- Orange Money API Documentation — Payment gateway integration reference.
- MTN Money API Documentation — Payment gateway integration reference.
- IEEE Std 830-1998 — IEEE Recommended Practice for Software Requirements Specifications.

1.5 Overview

This document is organized into the following sections: Section 2 provides an overall description of the product, including its perspective, functions, user characteristics, and constraints. Section 3 details the specific system features and functional requirements. Section 4 describes external interface requirements. Section 5 outlines non-functional requirements such as performance, security, and scalability. Section 6 contains appendices with supplementary information.

2. OVERALL DESCRIPTION

2.1 Product Perspective

AIDCELIX is a new, self-contained E-Pharmacy platform designed to bridge the gap between patients and local pharmacies. It operates as a centralized web and mobile application that aggregates pharmacy inventory data, medication pricing from MEDindex Pro, and delivery logistics through the Gozem API. The system does not replace existing pharmacy management systems but rather provides a consumer-facing interface for stock checking, price comparison, ordering, and delivery coordination.

2.2 Product Functions

ID	Function	Description
PF-01	User Account Management	Allow users to create and manage personal accounts.
PF-02	Location-Based Pharmacy Search	Identify user location and find nearest pharmacies with specific medications.
PF-03	Medication Search & Display	Search medications with images and pricing from MEDindex Pro.
PF-04	Medication Ordering	Place orders at chosen pharmacies with delivery via Gozem.
PF-05	Payment Processing	In-app payment via Orange Money, MTN Money, and other methods.
PF-06	Delivery Coordination	API integration with Gozem for delivery notification and tracking.
PF-07	Cross-Platform Access	Support for both mobile phones and computers.

2.3 User Classes and Characteristics

User Class	Description	Privileges
Registered User	Individual who has created a personal account on AIDCELIX.	Full access to search, order, pay, and track deliveries.
Guest User	Unregistered visitor browsing the application.	Limited to medication search and price viewing only.
Administrator	System administrator managing the platform.	User management, pharmacy onboarding, pricing oversight, system configuration.
Pharmacy Partner	Registered pharmacy using the platform to list inventory.	Manage stock, receive orders, update availability.

2.4 Operating Environment

The AIDCELIX application shall operate in the following environments:

Mobile Platforms: iOS (version 14.0 and above) and Android (version 8.0 and above).

Desktop Platforms: Modern web browsers including Google Chrome, Mozilla Firefox, Safari, and Microsoft Edge (latest two versions).

Server Environment: Cloud-based infrastructure with support for RESTful API services, database management, and real-time notifications.

Network: Internet connectivity required for all features; GPS/location services required for pharmacy proximity search.

2.5 Design and Implementation Constraints

- **Pricing Dependency:** Medication prices must be sourced from the external MEDindex Pro application/database. AIDCELIX must implement an additional 2% markup on all displayed prices.
- **Delivery Integration:** All delivery logistics are handled exclusively through the Gozem application via API integration. AIDCELIX cannot use alternative delivery providers without system modification.
- **Payment Gateway Limitations:** Primary payment methods are restricted to Orange Money and MTN Money, with provisions for additional mobile money services.
- **Regulatory Compliance:** The system must comply with local pharmaceutical regulations, data privacy laws, and mobile money transaction standards.
- **Cross-Platform Requirement:** The application must maintain functional parity between mobile and desktop versions.

2.6 Assumptions and Dependencies

- Users have access to a smartphone or computer with internet connectivity.
- MEDindex Pro maintains an up-to-date and accessible pricing database with API endpoints.
- Gozem delivery service operates in all target geographic areas and maintains API availability.
- Orange Money and MTN Money payment gateways are operational and accessible in the target market.
- Pharmacy partners are willing to integrate their inventory systems or manually update stock availability.
- Users grant necessary permissions for location services (GPS) on their devices.

3. SYSTEM FEATURES (FUNCTIONAL REQUIREMENTS)

3.1 User Account Management

FR-01 User Registration

Description: The system shall allow users to create a personal account by providing required information including name, email address, phone number, and a secure password. The system shall validate email uniqueness and phone number format. Upon successful registration, the user shall receive a confirmation notification.

Priority: High

Source Note: "In this APP user can create their personal account."

FR-02 User Authentication

Description: The system shall provide secure login functionality using registered email/phone number and password. The system shall support password recovery via email or SMS verification.

Priority: High

Source Note: "User account creation prerequisite."

FR-03 Profile Management

Description: The system shall allow registered users to view and update their profile information, including delivery addresses, contact details, and notification preferences.

Priority: Medium

Source Note: "Personal account management."

3.2 Location-Based Pharmacy Search

FR-04 Location Identification

Description: The system shall identify the user's current geographic location using GPS, Wi-Fi positioning, or manual address input. The system shall request location permissions and handle cases where location services are disabled by allowing manual address entry.

Priority: High

Source Note: "The APP should be able to identify the user location."

FR-05 Nearest Pharmacy Search

Description: The system shall display a list of registered pharmacies sorted by proximity to the user's location. For each pharmacy, the system shall display the name, address, distance, contact information, and operating hours.

Priority: High

Source Note: "Let the user know the nearest pharmacy that he or she can have a particular drug depending on his or her location."

FR-06 Medication Availability by Location

Description: The system shall allow users to search for a specific medication and display only those pharmacies within a defined radius that currently stock the requested drug. The system shall indicate stock availability status (In Stock, Low Stock, Out of Stock).

Priority: High

Source Note: "Nearest pharmacy that he or she can have a particular drug depending on his or her location."

3.3 Medication Search and Display

FR-07 Medication Search

Description: The system shall provide a search interface allowing users to search for medications by name, generic name, brand, or category. The search shall support autocomplete suggestions and handle spelling variations.

Priority: High

Source Note: "On the APP if the user search for any medication the app should display it."

FR-08 Medication Display with Images

Description: The system shall display medication search results with high-quality product images, medication name, dosage information, manufacturer details, and category classification.

Priority: High

Source Note: "The app should display it with images and prices of the medication."

FR-09 Pricing from MEDindex Pro

Description: The system shall retrieve real-time medication pricing from the external MEDindex Pro database via API integration. All displayed prices shall include an additional 2% markup calculated automatically by the AIDCELIX system. The original base price and the final price (base + 2%) shall be transparently displayed to the user.

Priority: High

Source Note: "The price of any medication should be prices found on the app name MEDindex Pro. The price of each medication should have an additional 2%."

3.4 Medication Ordering and Delivery

FR-10 Pharmacy Selection for Orders

Description: The system shall allow users to select their preferred pharmacy from the list of available pharmacies stocking the requested medication. Users shall be able to view pharmacy details, ratings, and estimated preparation time before confirming their selection.

Priority: High

Source Note: "The user should be able to pass commandes of his or her Medication in any pharmacy of her choice."

FR-11 Order Placement (Command)

Description: The system shall enable users to place an order (pass a command) for selected medications from their chosen pharmacy. The order process shall include a review step displaying medication details, quantities, unit prices, total medication cost, delivery fee, and final order total before confirmation.

Priority: High

Source Note: "Pass commandes of his or her Medication in any pharmacy of her choice."

FR-12 Delivery Service Integration

Description: The system shall coordinate delivery of ordered medications through integration with the Gozem delivery application. Upon order confirmation, the system shall automatically transmit delivery details (pickup pharmacy, delivery address, recipient contact, package details) to Gozem via API.

Priority: High

Source Note: "The delivery will be ensure by A delivery app name GOZEM."

FR-13 Delivery Fee Calculation

Description: The system shall calculate delivery fees based on real-time pricing returned by the Gozem API. The delivery fee shall vary depending on distance between the selected pharmacy and delivery address, package weight/volume, and Gozem's current pricing structure. The delivery fee shall be displayed to the user before order confirmation and is payable by the user.

Priority: High

Source Note: "If a user want to pass a Command the delivery is payable and the price for the delivery will depend on the Gozem app."

3.5 Payment Processing

FR-14 In-App Payment

Description: The system shall allow users to pay for their medication orders directly within the AIDCELIX application. Payment shall be processed after order confirmation and before pharmacy fulfillment. The system shall generate a digital receipt upon successful payment.

Priority: High

Source Note: "The APP should allow the user to be able to pay his or her medication on the app."

FR-15 Multiple Payment Methods

Description: The system shall support multiple mobile money payment methods including Orange Money and MTN Money as primary options. The system architecture shall be extensible to accommodate additional payment providers (e.g., Wave, PayPal, credit/debit cards) through a modular payment gateway integration pattern.

Priority: High

Source Note: "The different mode of payment are orange money, MTN Money and many other."

FR-16 Payment Security

Description: The system shall implement industry-standard encryption (SSL/TLS) for all payment transactions. Sensitive payment credentials shall never be stored on AIDCELIX servers; instead, tokenized references shall be used for transaction processing.

Priority: High

Source Note: "Secure payment processing requirement."

FR-17 Payment Confirmation

Description: The system shall provide immediate visual and notification-based confirmation of successful payment, including transaction ID, amount paid, timestamp, and payment method used. In case of payment failure, the system shall display clear error messages and allow retry or alternative payment method selection.

Priority: High

Source Note: "Payment completion workflow."

3.6 Gozem API Integration

FR-18 API Connection Establishment

Description: The system shall establish and maintain a secure API connection between AIDCELIX and the Gozem delivery application. The integration shall use authenticated API endpoints with proper error handling, retry logic, and timeout management.

Priority: High

Source Note: "Gozem APP and AIDCELIX will be connect by an API."

FR-19 Automatic Order Notification

Description: The system shall automatically notify the Gozem application each time a user places an order (passes a command) on AIDCELIX. The notification payload shall include all necessary delivery

information: pharmacy location, customer delivery address, contact details, medication list, package specifications, and preferred delivery time window.

Priority: High

Source Note: "Each time a user pass a command on Aidcelix app it should notify gozem app."

FR-20 Delivery Status Synchronization

Description: The system shall receive and display real-time delivery status updates from Gozem, including: Order Received, Driver Assigned, Pickup in Progress, En Route, and Delivered. Users shall receive push notifications and in-app updates at each status change.

Priority: Medium

Source Note: "API bidirectional communication."

3.7 Cross-Platform Compatibility

FR-21 Mobile Phone Support

Description: The system shall provide a fully functional native or responsive web application for mobile phones running iOS and Android. The mobile interface shall be optimized for touch interaction, smaller screen sizes, and mobile-specific features such as push notifications and GPS location services.

Priority: High

Source Note: "The should be use on both mobile phone."

FR-22 Computer/Desktop Support

Description: The system shall provide a fully functional web application accessible via desktop and laptop computers. The desktop interface shall be optimized for larger screens, mouse/keyboard interaction, and shall offer the same core functionality as the mobile version.

Priority: High

Source Note: "The should be use on both ... computers."

FR-23 Responsive Design

Description: The system shall implement responsive design principles to ensure usability across varying screen sizes and resolutions, from small mobile devices to large desktop monitors. All features, including search, ordering, payment, and delivery tracking, shall be accessible on all supported platforms.

Priority: Medium

Source Note: "Cross-platform consistency."

4. EXTERNAL INTERFACE REQUIREMENTS

4.1 User Interfaces

The AIDCELIX user interface shall adhere to the following guidelines:

- **Home Screen:** Display search bar, location indicator, nearby pharmacies preview, and quick-access categories.
- **Search Results Screen:** Display medication cards with image, name, price (base + 2% markup), and pharmacy availability.
- **Pharmacy Detail Screen:** Show pharmacy information, available medications, distance, ratings, and an 'Order Here' button.
- **Order Screen:** Display medication cart, pharmacy selection, delivery address input, delivery fee estimate from Gozem, and total cost breakdown.
- **Payment Screen:** Present payment method selection (Orange Money, MTN Money, Others), amount due, and secure payment form.
- **Order Tracking Screen:** Show real-time delivery status synchronized from Gozem API with map visualization where available.
- **Profile Screen:** Allow management of personal information, addresses, order history, and payment methods.

All interfaces shall follow accessibility standards (WCAG 2.1 Level AA) and support localization for target markets.

4.2 Hardware Interfaces

- **GPS/Location Services:** Interface with device GPS and network location services for accurate user positioning.
- **Camera:** Optional interface for scanning prescription QR codes or medication barcodes (future enhancement).
- **Biometric Sensors:** Support for fingerprint/face recognition for secure login on supported devices.
- **Storage:** Local caching of user preferences, search history, and temporary data for offline viewing of previously loaded content.

4.3 Software Interfaces

System	Interface Type	Data Exchanged	Protocol
MEDindex Pro	REST API	Medication names, images, base pricing, dosage info, manufacturer data	HTTPS/JSON
Gozem App	REST API / Webhooks	Delivery requests, pickup/dropoff addresses, package details, delivery status updates, pricing	HTTPS/JSON
Orange Money	Payment Gateway API	Payment requests, transaction status, confirmation codes, refunds	HTTPS/JSON
MTN Money	Payment Gateway API	Payment requests, transaction status, confirmation codes, refunds	HTTPS/JSON

Maps Service	SDK / API	Geocoding, routing, distance calculation, map tiles	HTTPS/JSON
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4.4 Communications Interfaces

- **Internet Connectivity:** All real-time features require active internet connection via Wi-Fi or mobile data.
- **Push Notifications:** Firebase Cloud Messaging (FCM) for Android and Apple Push Notification Service (APNS) for iOS to deliver order updates, delivery status, and promotional messages.
- **SMS Gateway:** Integration with SMS service provider for OTP verification, payment confirmations, and delivery notifications.
- **Email Service:** SMTP integration for account verification, order receipts, and marketing communications.

5. NON-FUNCTIONAL REQUIREMENTS

5.1 Performance Requirements

ID	Requirement	Target
NFR-01	Application launch time	≤ 3 seconds on mobile; ≤ 2 seconds on desktop
NFR-02	Medication search response time	≤ 2 seconds for results display
NFR-03	Pharmacy location query response	≤ 1.5 seconds for nearest pharmacy list
NFR-04	Payment transaction processing	≤ 10 seconds end-to-end including gateway response
NFR-05	Gozem API notification latency	≤ 5 seconds from order confirmation to Gozem receipt
NFR-06	MEDindex Pro price fetch	≤ 3 seconds with fallback to cached prices if unavailable
NFR-07	Concurrent user support	Minimum 10,000 simultaneous users without performance degradation
NFR-08	Page load time (desktop web)	≤ 1.5 seconds on standard broadband connection

5.2 Security Requirements

- **Data Encryption:** All data in transit shall be encrypted using TLS 1.3. Sensitive data at rest (PII, payment tokens) shall be encrypted using AES-256.
- **Authentication:** Multi-factor authentication (MFA) shall be available for user accounts. Passwords shall be hashed using bcrypt with salt.
- **Payment Security:** Compliance with PCI-DSS standards for payment processing. No raw payment card or mobile money PIN data shall be stored.
- **API Security:** All external API integrations (MEDindex Pro, Gozem, payment gateways) shall use OAuth 2.0 or API key authentication with request signing.
- **Input Validation:** All user inputs shall be validated and sanitized to prevent SQL injection, XSS, and CSRF attacks.
- **Session Management:** Secure session tokens with automatic expiration and invalidation on logout or suspicious activity detection.
- **Audit Logging:** All critical operations (login, payment, order placement, API calls) shall be logged with timestamps and user identifiers for forensic analysis.

5.3 Availability and Reliability

- **System Uptime:** The application shall maintain 99.9% uptime excluding scheduled maintenance windows.
- **Scheduled Maintenance:** Maintenance windows shall be announced 48 hours in advance and scheduled during off-peak hours (02:00 – 05:00 local time).
- **Disaster Recovery:** Automated daily backups with point-in-time recovery capability. Recovery Time Objective (RTO) of 4 hours and Recovery Point Objective (RPO) of 1 hour.
- **Failover:** Redundant server infrastructure with automatic failover for critical services (authentication,

payment, Gozem API relay).

- **Error Handling:** Graceful degradation when external services (MEDindex Pro, Gozem) are unavailable, with user-friendly error messages and cached data display where applicable.

5.4 Scalability

- **Horizontal Scaling:** The system architecture shall support horizontal scaling of application servers to handle increasing user load.
- **Database Scaling:** Database read replicas and sharding strategies shall be implemented to handle growth in medication catalog and transaction volume.
- **Geographic Expansion:** The system shall support deployment in new geographic regions with minimal configuration changes, including localization of currency, language, and pharmacy/delivery partner onboarding.
- **Feature Extensibility:** The codebase shall follow modular architecture principles to allow addition of new payment providers, delivery partners, and pharmacy integrations without major refactoring.

6. APPENDICES

Appendix A: Requirements Traceability Matrix

The following matrix maps each functional requirement to its source note from the original specification and identifies the responsible subsystem:

Req ID	Requirement Title	Source Note	Subsystem
FR-01	User Registration	User can create their personal account	User Management
FR-04	Location Identification	APP should be able to identify the user location	Location Services
FR-05	Nearest Pharmacy Search	Know the nearest pharmacy... particular drug depending on location	Pharmacy Finder
FR-07	Medication Search	User search for any medication the app should display it	Search Engine
FR-08	Medication Display with Images	Display it with images and prices	Search Engine
FR-09	Pricing from MEDindex Pro	Prices found on MEDindex Pro + additional 2%	Pricing / MEDindex Integration
FR-10	Pharmacy Selection for Orders	Pass commandes... in any pharmacy of her choice	Order Management
FR-12	Delivery Service Integration	Delivery ensured by GOZEM	Gozem Integration
FR-13	Delivery Fee Calculation	Delivery is payable... price depends on Gozem app	Gozem Integration
FR-14	In-App Payment	Pay medication on the app	Payment Gateway
FR-15	Multiple Payment Methods	Orange money, MTN Money and many other	Payment Gateway
FR-18	API Connection Establishment	Gozem and AIDCELIX connected by an API	Gozem Integration
FR-19	Automatic Order Notification	Each time user passes command, notify Gozem	Gozem Integration
FR-21	Mobile Phone Support	Use on mobile phone	Client Application
FR-22	Computer/Desktop Support	Use on computers	Client Application

Appendix B: Document Revision History

Version	Date	Author	Description of Changes
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1.0	August 22, 2026	Product Owner / Business Analyst	Initial draft based on handwritten specification notes. All core requirements from source material documented and expanded into formal SRS structure.
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Appendix C: Glossary of Terms

Term	Definition
Command	A user action to place an order for medication through the AIDCELIX application. Equivalent to 'order' or 'purchase request'.
E-Pharmacy	An electronic platform that enables online search, comparison, and ordering of pharmaceutical products.
Markup	The additional 2% added to the base medication price retrieved from MEDindex Pro, representing AIDCELIX's service fee.
Mobile Money	A digital payment service allowing financial transactions through a mobile device, prevalent in African markets (e.g., Orange Money, MTN Money).
Stock Checker	A system feature that verifies and displays real-time medication availability across registered pharmacy partners.

End of Document